

| Question |     |       |  | Marking details                                                                                                                                                                                                                     | Marks available |     |     |       |       |      |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                     | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 10/2     | (a) | (i)   |  | award (1) for any of following <ul style="list-style-type: none"><li>they have same number of electrons in their outer shell</li><li>they have 1 electron in their outer shell</li><li>they lose 1 electron when reacting</li></ul> | 1               |     |     | 1     |       |      |
|          |     | (ii)  |  | density                                                                                                                                                                                                                             |                 |     | 1   | 1     |       |      |
|          | (b) | (i)   |  | stored in oil / liquid paraffin<br><br>do not accept paraffin                                                                                                                                                                       | 1               |     |     | 1     |       | 1    |
|          |     | (ii)  |  | sodium hydroxide / NaOH (1)<br><br>hydrogen / H <sub>2</sub> (1)                                                                                                                                                                    | 2               |     |     | 2     |       |      |
|          |     | (iii) |  | lithium / Li                                                                                                                                                                                                                        | 1               |     |     | 1     |       |      |
|          |     | (iv)  |  | NaF                                                                                                                                                                                                                                 |                 | 1   |     | 1     |       |      |
|          | (c) | (i)   |  | 61 (2)<br><br>if incorrect award (1) for any of following figures in method or as final answer<br>114<br>0.61 / 0.606<br>10.1 / 10                                                                                                  |                 | 2   |     | 2     | 2     |      |
|          |     | (ii)  |  | mass medication / medical treatment without permission / no choice<br><br>neutral answers<br>any health problem<br>not needed because it's in toothpaste                                                                            | 1               |     |     | 1     |       |      |

| Question |  |       |  | Marking details                                                                                      | Marks available |     |     |       |       |      |
|----------|--|-------|--|------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |       |  |                                                                                                      | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  | (iii) |  | toothpaste<br><br>accept mouthwash<br><br>neutral answers<br>dental products<br>fluoride supplements | 1               |     |     | 1     |       |      |
|          |  |       |  | Question 10/2 total                                                                                  | 7               | 3   | 1   | 11    | 2     | 1    |